PS#11: Modeling with differential equations - Answer key

1. (AC 7.5 Exercise 7) When a skydiver jumps from a plane, gravity causes her downward velocity to increase at the rate of $g \approx 9.8$ m/s per second. At the same time, wind resistance causes her velocity to decrease at a rate proportional to the velocity.

   (a) Using k to represent the constant of proportionality, write a differential equation that describes the rate of change of the skydiver's velocity.

   $$\frac{dv}{dt} = 9.8 - k \cdot v.$$

   (b) Find any equilibrium solutions and decide whether they are stable or unstable. Your result should depend on k.

   $$\begin{aligned} 0 = \frac{dv}{dt} &= 9.8 - k \cdot v \\ k \cdot v &= 9.8 \\ v &= \frac{9.8}{k} \end{aligned}$$

   Since the phase graph crosses from above the v-axis to below the v-axis at this point, this is a stable equilibrium.

   (c) Suppose that the initial velocity is zero. Find the velocity $v(t)$.

   This equation is separable. First I'll factor out k.

   $$\begin{aligned} \frac{dv}{dt} = 9.8 - k \cdot v &= k \cdot \left(\frac{9.8}{k} - v \right) \\ \frac{1}{9.8/k - v}\, dv &= k\, dt \\ \int \frac{1}{9.8/k - v}\, dv &= \int k\, dt \\ -\ln\left(\frac{9.8}{k} - v \right) &= kt + C \\ \frac{9.8}{k} - v &= C \cdot \exp(-kt) \\ v(t) &= \frac{9.8}{k} - C \cdot \exp(-kt) \\ 0 = v(0) &= \frac{9.8}{k} - C \cdot \exp(-k \cdot 0) \\ C &= \frac{9.8}{k} \\ v(t) &= \frac{9.8}{k}\left(1 - \exp(-k \cdot t)\right) \end{aligned}$$

(d) A typical terminal velocity for a skydiver falling face down is 54 meters per second. What is the value of k for this skydiver?

The terminal velocity is the stable equilibrium solution:

$$54 = \frac{9.8}{k}$$
$$k = \frac{9.8}{54} \approx 0.1815$$

Therefore, $v(t) = 54(1 - \exp(-0.1815t))$.

(e) How long does it take to reach 50% of the terminal velocity?

$$\begin{aligned} 27 &= 54(1 - \exp(-0.1815t)) \\ 1/2 &= 1 - \exp(-0.1815t) \\ \exp(-0.1815t) &= 1/2 \\ -0.1815t &= \ln(1/2) = -\ln(2) \\ t &= \frac{\ln(2)}{0.1815} \approx 3.82 \text{ seconds.} \end{aligned}$$

2. (AC 7.5 Exercise 8) During the first few years of life, the rate at which a baby gains weight is proportional to the reciprocal of its weight.

(a) Express this fact as a differential equation.

Using $W(t)$ for the weight of the baby,

$$\frac{dW}{dt} = k \cdot \frac{1}{W}.$$

(b) Suppose that a baby weighs 8 pounds at birth and 9 pounds one month later. How much will he weigh at one year?

First let's separate and solve this differential equation:

$$\begin{aligned} \frac{dW}{dt} &= k \cdot \frac{1}{W} \\ W\, dW &= k\, dt \\ \int W\, dW &= \int k\, dt \\ \frac{W^2}{2} &= kt + C \\ W^2 &= 2kt + C \\ W(t) &= \sqrt{2kt + C} \end{aligned}$$

Now we can use the given data points to find the values of k and C. (Notice that t is measured in months.)

$$W(0) = 8 = \sqrt{2k \cdot 0 + C} \qquad W(1) = 9 = \sqrt{2k \cdot 1 + C}$$
$$8 = \sqrt{C} \qquad 9 = \sqrt{2k + 64}$$
$$C = 64 \qquad 2k = 81 - 64 = 17$$
$$k = 8.5$$

Therefore, $W(t) = \sqrt{17t + 64}$. To find the baby's weight at one year, we simply evaluate at $t = 12$ months:

$$W(12) = \sqrt{17 \cdot 12 + 64} = \sqrt{268} \approx 16.37 \text{ pounds.}$$

(c) Do you think this is a realistic model for a long time?

So, I mean, probably not, for several reasons:

- This function $W(t)$ is always increasing and has no horizontal asymptote; the model thus predicts that the person will keep growing forever.
- This model predicts a pretty low weight for someone who's 20 years old: $W(240) \approx 64.3$ pounds. That's a very small adult! I think this model doesn't know about various "growth spurts" that humans go through.

3. (AC 7.5 Exercise 9) Suppose that you have a water tank that holds 100 gallons of water. A briny solution, which contains 20 grams of salt per gallon, enters the tank at the rate of 3 gallons per minute. At the same time, the solution is well mixed, and water is pumped out of the tank at the rate of 3 gallons per minute.

(a) Since 3 gallons enter the tank every minute and 3 gallons leave every minute, what can you conclude about the volume of water in the tank?

The volume of water in the tank is in fact constant.

(b) How many grams of salt enter the tank every minute?

$$\frac{20 \text{ grams}}{\text{gallon}} \cdot \frac{3 \text{ gallons}}{\text{minute}} = 60 \frac{\text{grams}}{\text{minute}}.$$

(c) Suppose that $S(t)$ denotes the number of grams of salt in the tank in minute t. How many grams are there in each gallon in minute t?

$$\frac{S(t) \text{ grams}}{100 \text{ gallons}} = \frac{S}{100} \frac{\text{grams}}{\text{gallon}}.$$

(d) Since water leaves the tank at 3 gallons per minute, how many grams of salt leave the tank each minute?

$$\frac{S}{100} \frac{\text{grams}}{\text{gallon}} \cdot \frac{3 \text{ gallons}}{\text{minute}} = \frac{3S}{100} \frac{\text{grams}}{\text{minute}}.$$

(e) Write a differential equation that expresses the total rate of change of S.

$$\begin{aligned} \text{total rate} &= \text{rate in} - \text{rate out} \\ \frac{dS}{dt} &= 60 - \frac{3S}{100} \end{aligned}$$

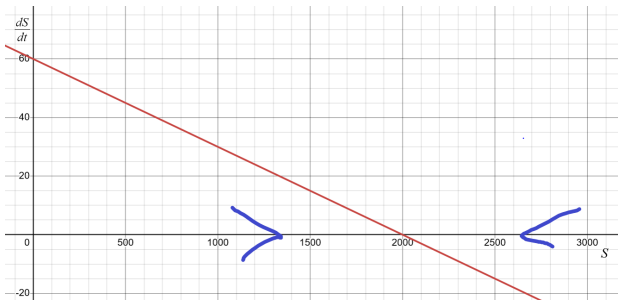
Notice that the units on everything here is $\frac{\text{grams}}{\text{minute}}$, so everything here is indeed a rate.

(f) Identify any equilibrium solutions and determine whether they are stable or unstable.

Let's see when $\frac{dS}{dt}$ is equal to 0:

$$\begin{aligned} 0 = \frac{dS}{dt} &= 60 - \frac{3S}{100} \\ \frac{3S}{100} &= 60 \\ 3S &= 6000 \\ S &= \frac{6000}{3} = 2000 \end{aligned}$$

Based on the phase graph, this is a stable equilibrium.



(g) Suppose that there is initially no salt in the tank. Find the amount of salt $S(t)$ in minute t.

Let's begin by separating and solving. I'm choosing to leave the $\frac{1}{100}$ on the right.

$$\begin{aligned} \frac{dS}{dt} &= \frac{1}{100}(6000 - 3S) \\ \int \frac{1}{6000 - 3S}\, dS &= \int \frac{1}{100}\, dt \\ -\frac{1}{3}\ln(6000 - 3S) &= \frac{1}{100}t + C \\ \ln(6000 - 3S) &= -\frac{3}{100}t + C \\ 6000 - 3S &= C \cdot \exp\left(-\tfrac{3}{100}t\right) \\ S(t) &= 2000 - C \cdot \exp\left(-\tfrac{3}{100}t\right) \\ 0 = S(0) &= 2000 - C \cdot \exp\left(-\tfrac{3}{100} \cdot 0\right) \\ C &= 2000 \\ S(t) &= 2000 - 2000 \cdot \exp\left(-\tfrac{3}{100}t\right) \end{aligned}$$

(h) What happens to $S(t)$ after a very long time? Explain how you could have predicted this only knowing how much salt there is in each gallon of the briny solution that enters the tank.

$S(t)$ rises toward the stable equilibrium of 2000 grams. This makes sense because the concentration of brine in the tank ($\frac{S(t)}{100}$ grams per gallon) will eventually match the concentration of brine flowing in (20 grams per gallon).